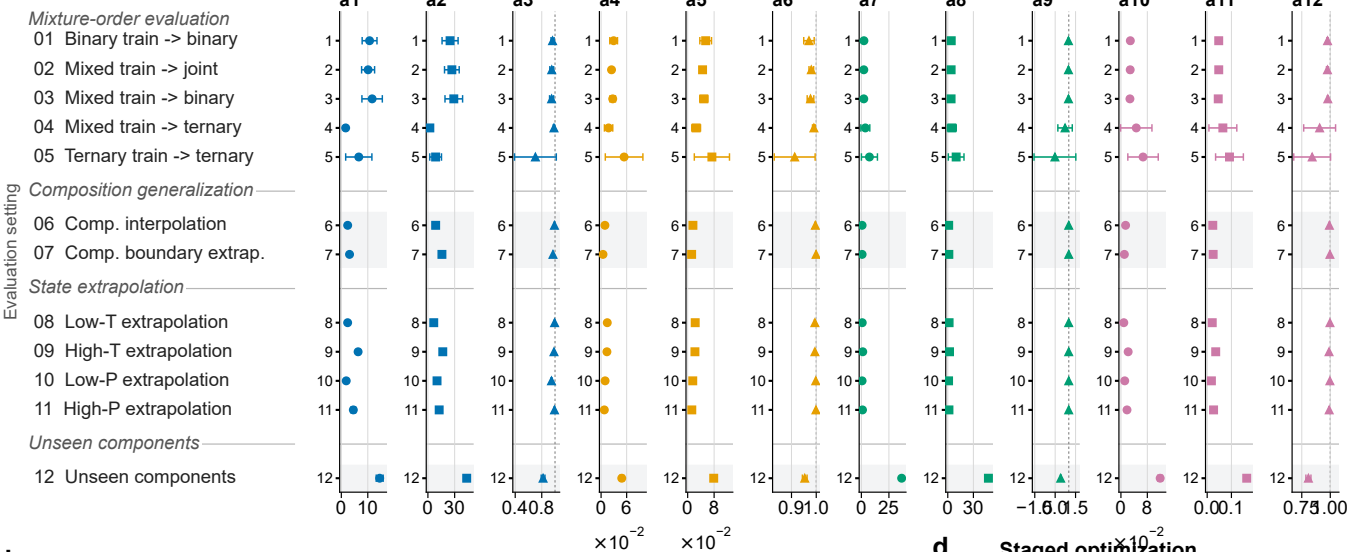
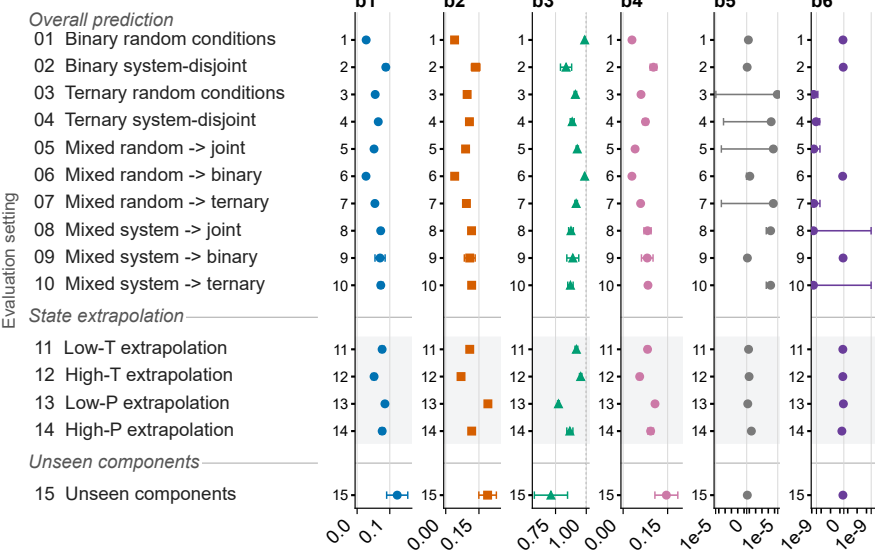


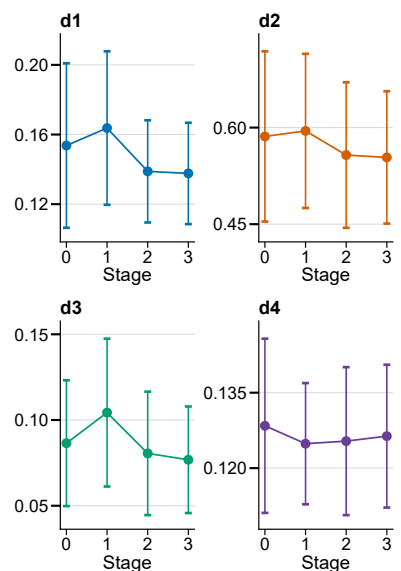
a VLE prediction and generalization



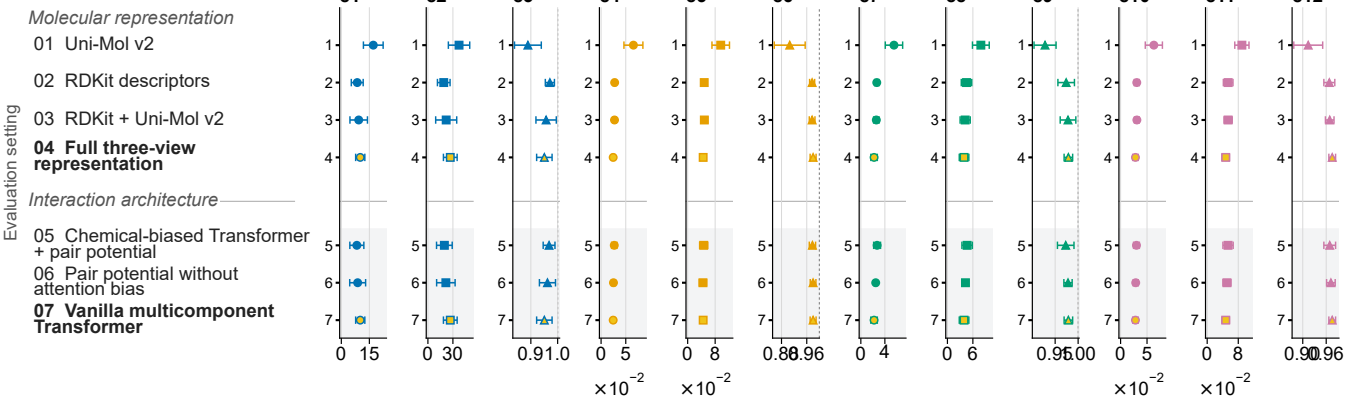
b LLE prediction and solver diagnostics



d Staged optimization



c VLE ablation



a/c: 1-3 Iso. P [MAE, RMSE, R2]; 4-6 Iso. y; 7-9 Isob. T; 10-12 Isob. y.

b: 1 MAE; 2 RMSE; 3 R2; 4 set distance; 5 activity RMS; 6 grid TPD. d: 1 mean $|r_f|$; 2 P95 $|r_f|$; 3 r_P ; 4 $\ln \gamma$ error.

Points show five-seed means; whiskers show sample SD. Symlog axes use a $\pm 10^{-14}$ linear region around zero.